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Jawaban

$$y_0 = -x^2 + 3x$$

$$y_1 = 2x^3 - x^2 - 5x$$

Pada $x \in [-2, 0]$, $y_0(x) \geq y_1(x) \Rightarrow y_0 - y_1$ $x \in [0, 2]$, $y_0(x) \leq y_1(x) \Rightarrow y_1 - y_0$

Maka, luas dipecah menjadi 2

$$A = \int_{-2}^0 y_1 - y_0 dx + \int_0^2 y_0 - y_1 dx$$

$$= \int_{-2}^0 2x^3 - 8x dx + \int_0^2 8x - 2x^3 dx$$

$$= \left. \frac{1}{2} 2x^4 - 4x^2 \right|_{-2}^0 + \left. 4x^2 - \frac{1}{2} 2x^4 \right|_0^2$$

$$= 8 + 8$$

$$= \boxed{16}$$

← cara menenguk x (karena y atas)

Dipecah menjadi dua bagian luas dengan

$$L_1' = x - \frac{x^2}{4}, \quad x \in [0, 1]$$

$$L_2' = 1 - \frac{x^2}{4}, \quad x \in [1, 2]$$

Gunakan integral

$$\begin{aligned} L_1 &= \int_0^1 L_1' dx = \int_0^1 x - \frac{x^2}{4} dx \\ &= \left. \frac{1}{2} x^2 - \frac{1}{12} x^3 \right|_0^1 \\ &= \frac{5}{12} \end{aligned}$$

$$\begin{aligned} L_2 &= \int_1^2 L_2' dx = \int_1^2 1 - \frac{x^2}{4} dx \\ &= \left. x - \frac{1}{12} x^3 \right|_1^2 \\ &= \left(2 - \frac{8}{12} \right) - \left(1 - \frac{1}{12} \right) \\ &= \frac{5}{12} \end{aligned}$$

$$L_1 + L_2 = \boxed{\frac{5}{6}}$$

Cara II. tingas dan sunby
atas $y = \frac{x^2}{4}$

$$\Rightarrow x = 2\sqrt{y}$$

$$\text{bawah} \Rightarrow y = x$$

$$\Rightarrow x = y$$

y dari 0 sampai 4, maka

$$\int_0^4 2\sqrt{y} - y \, dy$$

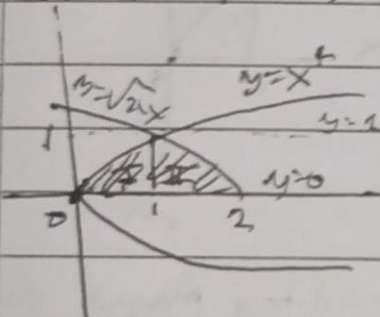
$$= \left[\frac{4}{3} \sqrt{y}^3 - \frac{1}{2} y^2 \right]_0^4$$

$$= \frac{5}{6}$$

3 $x = y^4$ dan $y = \sqrt{2-x}$

$y = \sqrt{x}$ dan $y = \sqrt{2-x}$

Sketsa kasar:



Cara I

pecah pakai integral biasa

$$\Rightarrow L_I = \int_0^1 \sqrt{x} \, dx$$

$$L_{II} = \int_1^2 \sqrt{2-x} \, dx$$

$$\Rightarrow L_I = \left[\frac{2}{5} \sqrt{x}^5 \right]_0^1 + \left[-\frac{2}{3} \sqrt{(2-x)^3} \right]_1^2$$

$$= \frac{2}{5} + \left(0 + \frac{2}{3} \right)$$

$$= \frac{22}{15}$$

Cara II

Tinjau berdasarkan sumbu y

$$x = y^4 \text{ dan } x = 2 - y^2$$

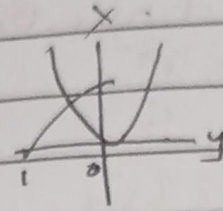
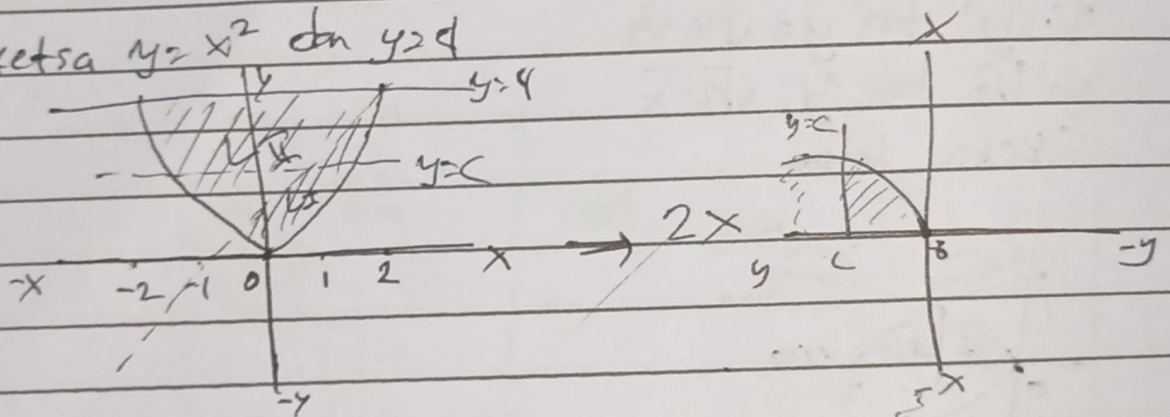
dari $y = 0$ sampai $y = 1$

$$\int_0^1 ((2 - y^2) - y^4) dy$$

$$= 2y - \frac{1}{3}y^3 - \frac{1}{5}y^5 \Big|_0^1$$

$$= 2 - \frac{1}{3} - \frac{1}{5}$$

$$= \frac{22}{15}$$

Sketsa $y = x^2$ dan $y = 4$ 
 $L_I = L_{II}$, garis dengan $y = c$ agar d. tinggi menjadi dua

maka

$$L_{cx^2} = \frac{1}{2} L_{(-2,2)} \\ \Rightarrow 2 \int_0^c \sqrt{y} dy = \frac{1}{2} \int_{-2}^2 x^2 dx$$

$$\Rightarrow \frac{2 \cdot 2\sqrt{c^3}}{3} = \left(\frac{1}{2} \cdot 2 \cdot \frac{1}{3} x^3 \Big|_{-2}^2 \right) + 8$$

$$\Rightarrow \frac{4\sqrt{c^3}}{3} = \frac{8}{3} + 8$$

$$\Rightarrow c^3 = 16$$

$$\Rightarrow \underline{c = \sqrt[3]{16}}$$

$$\frac{1}{2} \int_{-2}^2 x^2 dx = \frac{1}{2} \left(\frac{1}{3} x^3 \Big|_{-2}^2 \right) = \frac{1}{2} \left(\frac{8}{3} - \left(-\frac{8}{3} \right) \right) = \frac{1}{2} \left(\frac{16}{3} \right) = \frac{8}{3}$$

$$\Rightarrow \frac{4\sqrt{c^3}}{3} = \frac{8}{3} + 8$$

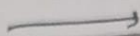
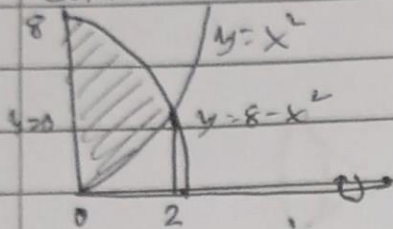
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R: CQI, $y = 8 - x^2$, $y = x^2$, $y = 0$

Volume pejal mengitari

(a)

Sumbu-x



$$y_0 = y_1$$

$$\Rightarrow x^2 = 8 - x^2 \Rightarrow x = 2 \text{ (ambil positif : QI)}$$

Maka, Volumena

$$V = \pi \int_0^2 (8 - x^2)^2 - (x^2)^2 dx$$

$$= \pi \int_0^2 (64 - 32x^2) dx$$

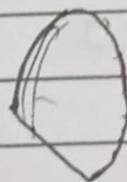
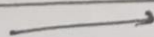
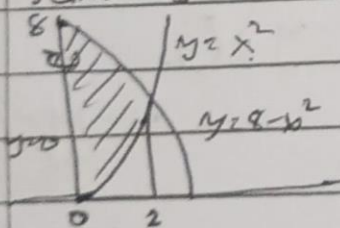
$$= \left. 64x - \frac{32x^3}{3} \right|_0^2 \cdot \pi$$

$$= \left(128 - \frac{32 \cdot 2^3}{3} \right) \cdot \pi$$

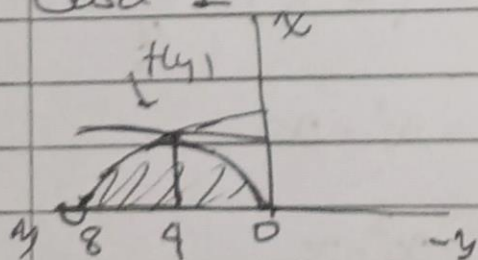
$$= \frac{256\pi}{3}$$

(b)

Sumbu-y



Cara I

Simetris di/pada $y \in [8, 9]$ (pembuktian dapat mengecek bahwa $f(y-4)$ itu genap)

Dengan demikian, akan digunakan interval pada $[0, 4]$ dengan $f(y) = \sqrt{y}$, kemudian dikali dua

$$V = \pi \int_0^4 y \, dy \cdot 2$$

$$= 2\pi \left. \frac{1}{2} y^2 \right|_0^4$$

$$= \underline{16\pi}$$

Cara II

Pakai bawang/kulit tabung

$$f(x) = (8 - x^2) - x^2$$

$$= 8 - 2x^2$$

Maka

$$V = \int_0^2 2\pi x (8 - 2x^2) \, dx$$

$$= 2\pi \int_0^2 8x - 2x^3 \, dx$$

$$= 2\pi \left(4x^2 - \frac{1}{2} x^4 \right) \Big|_0^2$$

$$= 2\pi (16 - 8)$$

$$= \underline{16\pi}$$

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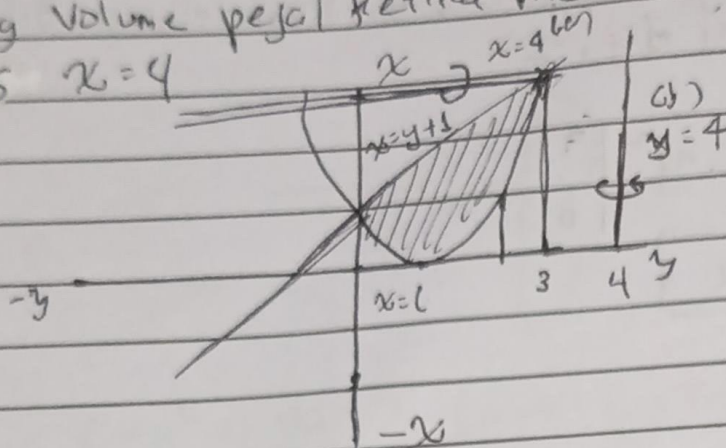
R dibatasi kurva-kurva $x = (y-1)^2$ dan $x - y = 1$

Let $x = y + 1$

Hitung Volume pejal ketika memutar

(a)

garis $x = 4$



(b)

$x = y + 1$ "bawah"

$x = (y-1)^2$ "atas"

↳ karena lebih jauh

dari $x = y + 1$

terhadap $x = 4$

Menggunakan metode cincin/washer

$$V = \pi \int_1^4 (4 - (y-1)^2) - (4 - (y+1)) dy$$

$$= \pi \int_1^4 ((-y^2 + 2y + 3)^2 - (3-y)^2) dy$$

$$= \pi \int_1^4 (y^4 - 4y^3 - 3y^2 + 18y) dy$$

$$= \pi \left(\frac{1}{5} y^5 - y^4 - y^3 + 9y^2 \right) \Big|_1^4$$

$$= \frac{108\pi}{5}$$

(b)

garis $y = 4$

Menggunakan metode kulit tabung

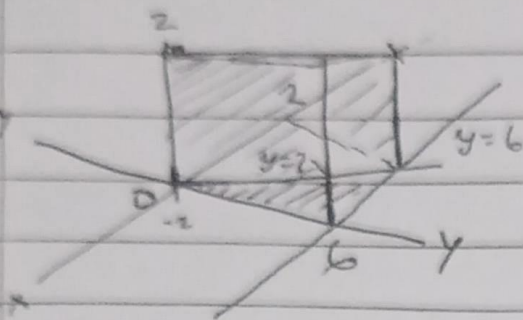
$$V = 2\pi \cdot r \cdot h$$

$$= \int_0^3 2\pi (4-y) \cdot ((y+1) - (y-1)^2) dy$$

$$= \int_0^3 2\pi (y^3 - 7y^2 + 12y) dy$$

$$= 2\pi \left(\frac{1}{4} y^4 - \frac{7}{3} y^3 + 6y^2 \right) \Big|_0^3$$

$$= \frac{45\pi}{2}$$



$$y_1 = y_2$$

$$\Rightarrow 6 = 3x$$

$$\Rightarrow x = 2$$

Tidak perlu memakai integral

(Meskipun jawabannya sama)

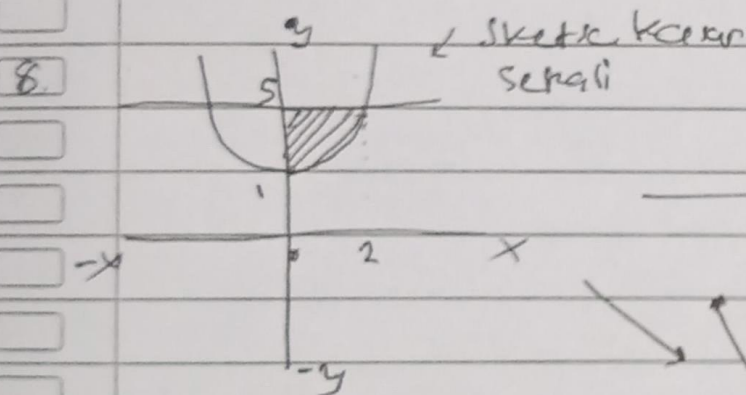
Perhatikan bahwa alasnya berbentuk segitiga sehingga

$$A = \frac{x_{\max} y_{\max}}{2} = \frac{6 \cdot 2}{2} = 6$$

$$V = At$$

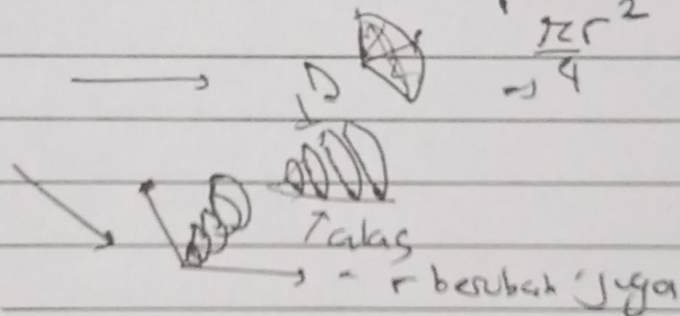
$$= 6 \cdot 10$$

$$= \frac{60}{2}$$



Perhatikan

$$\frac{\pi r^2}{4}$$



$$r = x = \sqrt{y-1} \quad (\text{karena terhadap sumbu } y)$$

$$\begin{aligned} \Rightarrow V &= \int_1^5 \frac{\pi}{4} \cdot r^2 \, dy \\ &= \frac{1}{4} \pi \int_1^5 (\sqrt{y-1})^2 \, dy \\ &= \frac{1}{4} \pi \left(\frac{1}{2} y^2 - y \right) \Big|_{y=1}^{y=5} \\ &= \frac{2\pi}{2} \end{aligned}$$